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Portfolio Construction & Portfolio Performance

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Executive Summary

The report discusses the construction of a portfolio that maximized the risk-adjusted return in 2017, whose task was given by the CEO of our Wall Street company. Apart from achieving a high possible return, the focus of the portfolio is that they should contain large companies listed on the New York stock exchange and none of them can be weighted larger than 10% in the portfolio. As we are only considering long positions, oil and gas companies are limited in the portfolio because of their potentially high volatility.

After the portfolio was constructed in 2017 (Portfolio A1), it is assumed that our clients invested in this portfolio during 2012-2017. Hence, a portfolio performance in terms of return & risk was analyzed against its benchmark, the S&P500. In the latter period (Portfolio A2), 2018-2022, assumed that the portfolio had been invested the same as in 2017. An evaluation of the portfolio performance was also constructed for this period. However, through further analysis, a different portfolio is recommended with the goal of maximizing return for a given level of risk (Portfolio B). Below is the summary of all 3 portfolios (Figure a).

COMPANY	WEIGHT		
	PORTFOLIO A1 (2012-2017)	PORTFOLIO A2 (2018-2022)	PORTFOLIO B (2018-2022)
UNITEDHEALTH GROUP	10%	10%	10%
INTERNATIONAL BUS.MCHS.	0%	0%	0%
JOHNSON & JOHNSON	10%	10%	0%
WALMART	10%	10%	10%
JP MORGAN CHASE & CO.	0%	0%	0%
P&G	0%	0%	10%
VISA	10%	10%	0%
ELI LILLY	10%	10%	10%

HOME DEPOT	10%	10%	0%
MASTERCARD	9.52%	9.52%	10%
BANK OF AMERICA	8.44%	8.44%	0%
COCA COLA	0%	0%	4.63%
PFIZER	0%	0%	7.47%
MERCK & COMPANY	2.04%	2.04%	10%
MORGAN STANLEY	0%	0%	0%
THERMO FISHER SCIENTIFIC	10%	10%	10%
CITIGROUP	0%	0%	5.59%
NEXTERA ENERGY	10%	10%	10%
EXXON MOBIL	0%	0%	2.32%
CHEVRON	0%	0%	0%
Total	100%	100%	100%

Figure a. Weight comparison of portfolio A1 (2017), portfolio A2 (2018-2022), and suggested portfolio B (2018-2022).

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Introduction

Choosing a good asset allocation among an enormous range of assets now can be facilitated if we consider the return risk ratio of a particular portfolio. A comparison between different portfolios in terms of return/risk performance can tell us, for instance, whether it is best to select one given that the portfolio has performed a Markowitz optimal procedure. Given the idea, it is likely more of interest for the investor to develop and estimate a maximum Sharpe Ratio portfolio rather than a minimum variance portfolio. Of course, risk-averse individuals would be more willing to expect the lowest risk for their investment portfolio. This process, however, would not promise a good plausible return from the market and is least favored by risk-taking individuals.

Understandably, some prospective clients that choose to invest in our company's fund may want their portfolio to outperform the overall market but not include any risk elements. Unfortunately, risk and return in the stock market have a positive relationship. The higher the return, the larger the risk. With the Markowitz optimization tool, an optimal portfolio is constructed by promising a maximum return with the least amount of risk, measured by the Sharpe ratio. Therefore, the higher the Sharpe ratio of an investment portfolio, the better the risk-adjusted return.

Methods

Monthly data of the period 12.01.2012-12.01.2017 & 01.01.2018-01.11.2022 for companies' total return index, S&P 500 price index, and US Treasury Bond 10 years Yield were collected using DataStream in Excel. The total return index has the advantage of assuming that all cash distribution is reinvested, and additionally tracks equity's price movements. The price index only considers the movement of securities, for instance, capital gains/losses of the index. The US long-term bonds (between 10-30 years) do carry some price risk in terms of inflation. As the investment period of our analysis is 5 years, it is still considered a relatively long-term investment. Hence, we think that the 10-year bond rate might be a good proxy in consideration for more market risk. However, due to the limitation of data collection, we only collected the price index for the benchmark while acknowledging that the total return index may have been an even closer fit for the analysis.

Companies' return is calculated using simple return. Max Sharpe's portfolio was constructed using Markowitz's portfolio optimization theory. T-statistics for Beta-value have been recalculated and tested for a value different from 1, instead of assuming 0 from the regression. All the significant results are shown in "grey" text.

The last suggested portfolio is constructed using Max Sharpe's optimizing theory. The argument behind the strategy is answering the following question: "What portfolio would have yielded the best risk-adjusted return for the period of 2018-2022?"

Portfolio Description

The portfolio contains 20 large companies listed on the New York Stock Exchange (Figure 1). They are chosen among 26 companies selected from DataStream in consideration of their size. Large companies typically have a large cap of more than \$10 billion (Fernando, 2022). Companies' market cap value is collected from Morningstar (Morningstar, Inc., 2022). The goal of constructing this portfolio is to maximize risk-adjusted return, hence, having a diversified portfolio is indeed needed. Therefore, 20 stocks are chosen given that the more different stocks in the portfolio, the less unsystematic risk the portfolio might be exposed to (Beers, 2022). There are 10 different sectors the portfolio will be investing in, including Insurance, Technology, Retail, Beverage, Consumer Goods, Financial Services, Pharmaceuticals, Biotechnology, Renewable Energy, and Oil and Gas. Financial sectors and Pharmaceuticals dominate half of the portfolio. Financial sectors tend to pay dividends to investors as they demonstrate great history, which makes them potentially great investments (Dhir, 2021). Giant pharmaceutical corporations have been seen as quite a solid place to invest, despite the debate of consumerism which is broad and indeed is not covered in the paper (Zucchi, 2021).

COMPANY	INDUSTRY	MARKET CAP (07.12.2022)
UNITEDHEALTH GROUP	INSURANCE	\$503.9131B
INTERNATIONAL BUS.MCHS.	TECHNOLOGY	\$133.3586 B
WALMART	RETAIL	\$404.2234 B
HOME DEPOT	RETAIL	\$323.5406 B
COCA COLA	BEVERAGE	\$274.3471 B
P&G	CONSUMER GOODS	\$353.7483 B
JP MORGAN CHASE & CO	FINANCIAL SERVICES	\$385.9798 B
VISA	FINANCIAL SERVICES	\$442.7518 B
MASTERCARD	FINANCIAL SERVICES	\$334.1361 B
BANK OF AMERICA	FINANCIAL SERVICES	\$264.7403 B
MORGAN STANLEY	FINANCIAL SERVICES	\$147.9015 B

CITIGROUP	FINANCIAL SERVICES	\$87.1196 B
ELI LILLY	PHARMACEUTICAL	\$350.1786 B
JOHNSON & JOHNSON	PHARMACEUTICAL	\$460.4106 B
PFIZER	PHARMACEUTICAL	\$279.0379 B
MERCK & COMPANY	PHARMACEUTICAL	\$276.1807 B
THERMO FISHER SCIENTIFIC	LABORATORY EQUIPMENT, BIOTECHNOLOGY	\$215.9154 B
NEXTERA ENERGY	RENEWABLE ENERGY	\$169.2666 B
EXXON MOBIL	OIL AND GAS	\$427.8083 B
CHEVRON	OIL AND GAS	\$332.6052 B

Figure 1. List of 20 companies in the portfolio, their respective industry, and market capitalization.

Portfolio Construction (2017)

Evaluation of Companies' Return and Risk

Before constructing a max Sharpe portfolio, it is essential to have a brief look at the performance of these companies in the 2012-2017 period. Doing this allows our clients to understand the reason behind their weight allocation in the portfolio, whether one should be weighted more or less than the others.

Figure 2 provides an overall annual return and annual risk of 20 companies in the portfolio. Generally, half of them perform quite well, more than 20%, relative to their risk. Companies with a higher return tend to enhance more volatility than others. However, many companies experience high losses, given that their return was relatively low compared to their risk. IBM seems to be the riskiest company, whose equity has a risk of 18 times larger than its relative return. An investor that expects a higher risk-adjusted return may not be very interested in the company. Conversely, UnitedHealth Group might promise a double return on its risk, which then becomes potential for our portfolio.

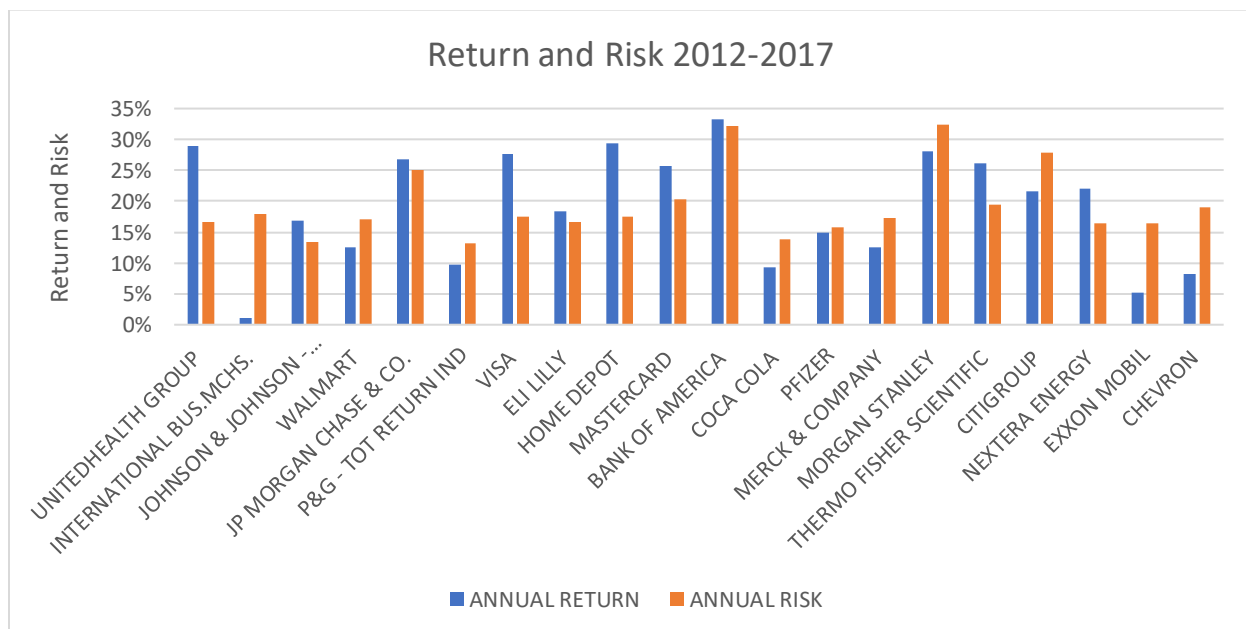


Figure 2. Companies' annual return and risk in 2012-2017.

Portfolio weight

Given a brief presentation of companies' return and risk, the max Sharpe portfolio is constructed. Max Sharpe's portfolio aims at achieving a high return as much as possible, given an acceptable risk. Compared to a Min Variance Portfolio, which minimizes the risk of investment as much as possible given a low level of return, the Max Sharpe portfolio is more desired by our clients.

With the restriction put on the portfolio before construction, almost half of all companies are given a weight of approximately 10%. Most of these companies have a higher return relatively than the risk they took, despite the potentially high volatility in this period. High volatility could be explained by the time horizon of the portfolio, hence, companies that are given weight in the portfolio potentially perform better than others.

As expected, IBM did not receive any weight in the portfolio due to its bad return performance. Other companies that are given 0% weight represented by the energy sector generally perform quite badly during 2012-2017. Among the 3 energy companies listed in the portfolio, only NEXTERA has been given a weight of 10%, despite the energy sector having a maximum restriction of 25% total in the portfolio. EXXON MOBIL and CHEVRON experience large losses, which does not seem plausible for achieving return compensation.

In the portfolio, half of the financial companies are given weights, such as VISA, MASTERCARD, and BANK OF AMERICA, given their return is relatively high. Other sectors such as insurance, retail, pharmaceutical, and biotechnology are together added up to 72% of the total weight. This makes the Max Sharpe portfolio quite diversified.

COMPANY	RETURN	RISK	WEIGHT
UNITEDHEALTH GROUP	29%	17%	10%
INTERNATIONAL BUS.MCHS.	1%	18%	0%
JOHNSON & JOHNSON	17%	13%	10%
WALMART	13%	17%	10%

JP MORGAN CHASE & CO.	27%	25%	0%
P&G	10%	13%	0%
VISA	28%	17%	10%
ELI LILLY	18%	17%	10%
HOME DEPOT	29%	18%	10%
MASTERCARD	26%	20%	9.52%
BANK OF AMERICA	33%	32%	8.44%
COCA COLA	9%	14%	0%
PFIZER	15%	15%	0%
MERCK & COMPANY	12%	17%	2.04%
MORGAN STANLEY	28%	32%	0%
THERMO FISHER SCIENTIFIC	26%	20%	10%
CITIGROUP	22%	28%	0%
NEXTERA ENERGY	22%	16%	10%
EXXON MOBIL	5%	16%	0%
CHEVRON	8%	19%	0%
Total			100%

Figure 3. Max Sharpe Portfolio weight.

Max Sharpe Portfolio Performance vs Benchmark

Assuming that the portfolio had been invested from 2012 to 2017, one might expect to know how the portfolio generated return relative to the return from its benchmark, S&P500. Figure 4 briefly shows the monthly simple return of our constructed portfolio, whose return seemed to be

slightly over the S&P500 within the period. Despite having a generous number of positive returns, some months experienced low and negative returns for the portfolio and its benchmark. However, one might expect the returns to fluctuate quite frequently from 0 because the data is displayed as monthly.

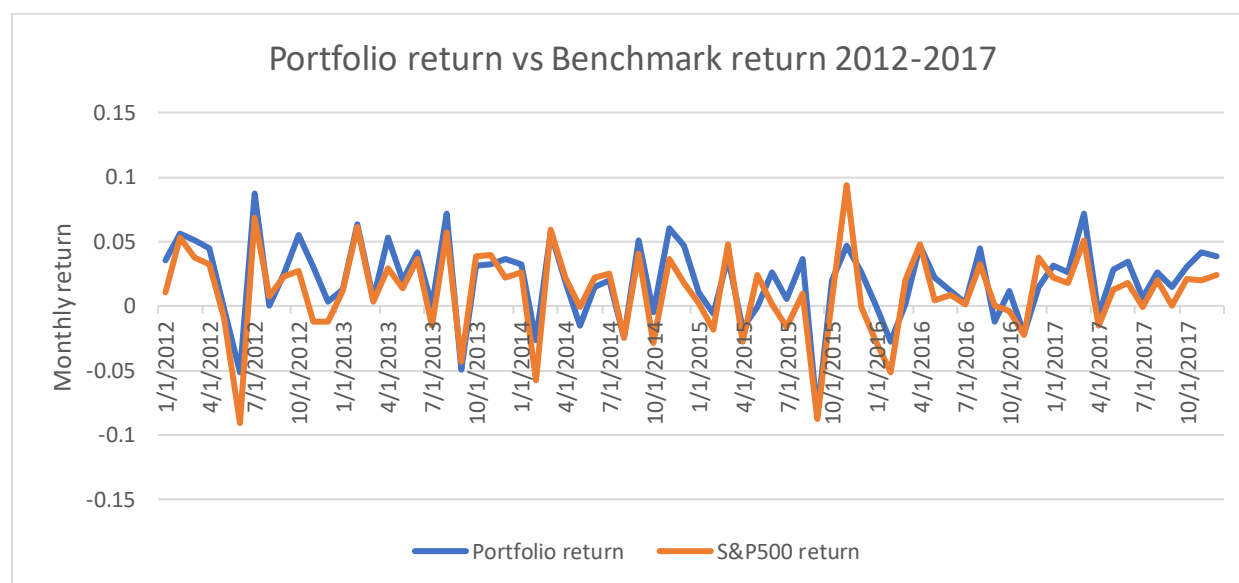


Figure 4. Monthly simple return of the Portfolio VS Benchmark during 2012-2017.

Our constructed portfolio is given as “Portfolio A1” in Figure 5. Portfolio A1 seemed to perform better than the benchmark, given a 23.3% return concerning 11% risk. With a T-value of 4.85, we’re confident to reject the null hypothesis that the return of our max Sharpe portfolio is equal to the return of the benchmark at the 5% level. Additionally, a 28% level of confidence is needed to reject the null hypothesis that the risk of Portfolio A1 is similar to the benchmark’s risk. This strongly supports the fact that our portfolio had beaten the Benchmark, giving the same level of risk the S&P500 was holding.

Portfolio A1 had a higher Sharpe ratio than its relative benchmark, 2.04 compared to 0.98, indicating that it had a higher level of risk-adjusted return than the S&P500. For an additional unit of risk-taking, our portfolio gained two more units of excess return if we did not invest in

the bonds market. A high Sharpe Ratio (>1) is usually desired by risk-taking investors when one would expect more return in investing.

The two portfolios were both negatively skewed and have a Platykurtic distribution, negative skewness indicates that the investors might expect frequent small gains as well as a few large losses. Platykurtic distribution has lighter tails and signals a moderate level of risk. This, unfortunately, is not supported by our data, given a fairly high volatility level.

2012-2017	Portfolio A1	S&P500
Annual return	23.8%	13.25%
Annual Risk	10.58%	11.34%
Risk-free rate	2.11%	2.11%
Sharpe Ratio	2.04	0.98
Kurtosis	1.21	1.54
Skewness	-0.67	-0.67
T-value	4.85	
P(T $\leq$ t) one-tail	3E-06	
F-value	0.87	
P(T $\leq$ t) one-tail	0.28	

Figure 5. Return/Risk Performance of The Max Sharpe portfolio VS Benchmark during 2012-2017.

Portfolio Performance (2018-2022)

Given that the portfolio constructed in 2017 has been invested the same during 2018-2022, we want to assess the portfolio's monthly return against its benchmark. Notice here that the S&P 500 has had a slightly more return than our portfolio in this period compared to 2017. However, both portfolios experienced a big dip in March 2020 due to Covid-19 and deviated a lot from 0 since the first quarter of 2022. The fluctuation from 2020 can be explained by market turbulence according to the global energy crisis and inflation. When the market went down by 10%, our portfolio lost approximately 5%.

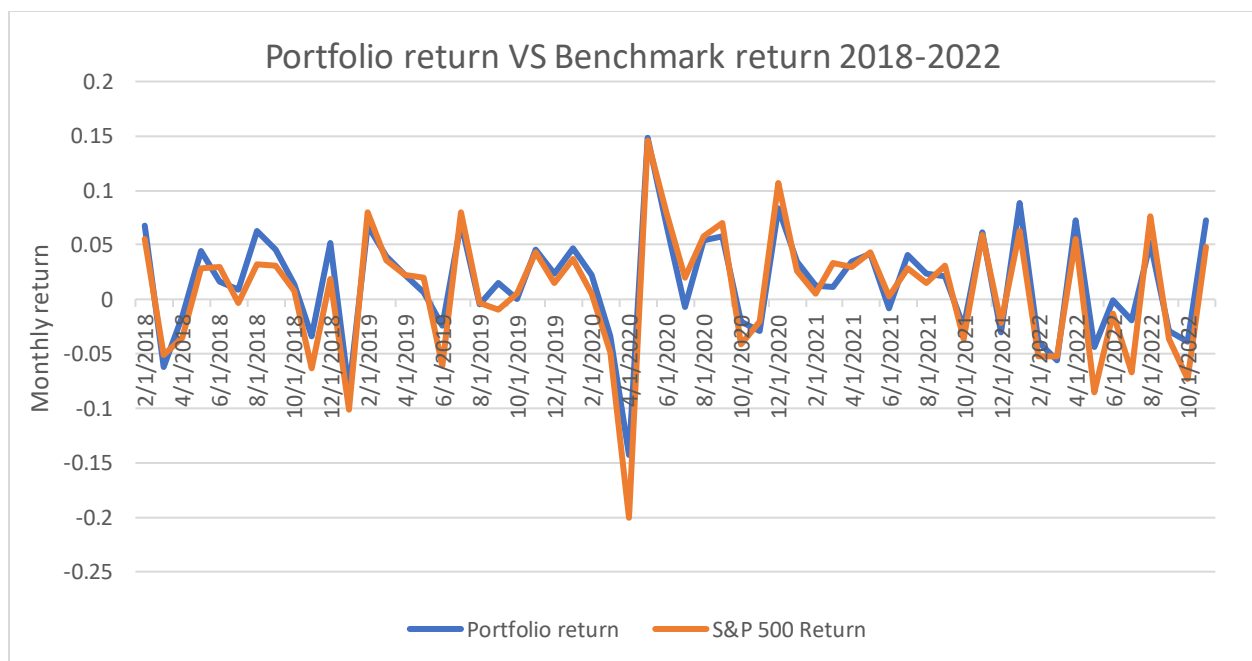


Figure 6. Monthly simple return of Portfolio VS Benchmark during 2018-2022.

2018-2022	Portfolio A2	S&P500
Annual return	18.69%	9.60%
Annual risk	17.03%	19.88%
Risk-free rate	2.41%	2.41%
Sharpe Ratio	0.96	0.36
Tracking Error	6.32%	
Information Ratio	1.44	
Treynor Ratio	0.20	0.07
Beta	0.82	1
T-value ($\beta = 1$)	-5.33	
Alpha	0.10	0
T-value ($\alpha = 0$)	4.42	
R square	0.91	

Figure 7. Portfolio performance against the Benchmark during 2018-2022.

The portfolio that has been investing the same as in 2017 is given by “Portfolio A2” in Figure 7. Overall, our portfolio seemed to outperform the benchmark again, supported by having a higher Sharpe ratio. The S&P 500 did not seem to perform well compared to our portfolio, reflected by having a lower Sharpe ratio. That explains a lower additional excess return for a given level of risk-taking by investing in the S&P500. The shortcoming of the Sharpe ratio is that it compares the portfolio’s total risk-adjusted return to the risk-free rate. As an investor, one might expect a better index for comparing returns, as these indices usually promise a higher return than the risk-free rate. Hence, other performance measures are introduced.

Tracking error (TE), generally, measures how a portfolio or a fund is managed and its corresponding volatility level. A higher TE indicates that the portfolio/fund has been actively taking risks against its benchmark. However, our portfolio was not exposed to much more risk than the S&P500, reflected by deviating from the benchmark of 6%. Information ratio (IR) is normally analyzed together with TE, whose TE is a component in calculating the IR. Our

portfolio had a good IR of 1.44, which indicates that it is more consistent in generating excess return compared to the benchmark.

Treynor's Ratio (TR) is another risk-adjusted return measurement. While the Sharpe ratio measures excess return in terms of portfolio risk, Treynor's ratio measures return relative to the portfolio's systematic risk (β). TR tells by how much additional return an investor might expect to gain per unit of market risk exposure. Our portfolio had a slightly higher TR than the benchmark, evidence that our portfolio outperformed the S&P 500 if adjusted for systematic risk. As our portfolio has taken less systematic risk than the benchmark, reflected by a Beta < 1, one would argue for a higher TR.

Overall, portfolio A2 has been able to beat the benchmark in the 2018-2022 period. This is also proven by having a high and significant Alpha, at 0.1. A positive Alpha shows that the portfolio's average return is above what is predicted by the CAPM, given its level of volatility. In other words, by how much our portfolio outperforms the market adjusting for systematic risk. Beta was also significant and relatively high for portfolio A2, showing that a unit increase in the benchmark's return would increase the portfolio's return by 0.82. Beta also explains that the portfolio has taken a slightly lower risk than the benchmark, supported by a 17% annual risk compared to the 20% risk of the benchmark. A High R square value indicated that more than 90% level of our portfolio's risk is explained by the S&P500. Because our relatively diversified portfolio helps eliminate its idiosyncratic risk, we would expect a lower risk than the benchmark despite having a high R square value.

Further Analysis

Overall, if investors were to invest in the same portfolio of 2017 in the 2018-2022 period (portfolio A2), one might expect the portfolio to beat the benchmark, proven by the previous analysis. However, to maximize risk-adjusted return, we suggest a different portfolio (Portfolio B) for the given period that would have yielded better performance. The portfolio is constructed by maximizing the Sharpe ratio for the period of 2018-2022 to obtain the highest possible return.

2018-2022	Portfolio A2	Portfolio B (suggested)
Annual return	18.68%	21.15%
Annual risk	17.03%	17.65%
Risk-free rate	2.41%	2.41%
Sharpe	0.96	1.06

Figure 8. Performance comparison between the 2018-2022 portfolio vs the suggested portfolio

In terms of return and risk, portfolio B promises a much higher return if one were to invest. Given the same level of volatility that the two portfolios have taken, portfolio B also has a better risk-adjusted excess return. With a unit of risk-taking, it can generate a unit of return in compensation for its risk (1:1). While portfolio A seems to outperform the benchmark during this period, it might not be plausible to invest as the portfolio of 2017 during the latter period. Hence, we provide an alternative portfolio as below (Figure 9).

Notice that we decided to keep the same restriction when constructing portfolio B. That is every company's weight must not exceed 10% and all energy company weight allocation sums up to 25%. Portfolio B has only 8 companies that are weighted by 10%, compared to ½ the number of companies having 10% in portfolio A2, but generally allocates its weight to slightly more companies. Companies that have their weight changed are highlighted as Grey in Figure 9. Johnson & Johnson, VISA, and Home Depot are to be weighted down to 0% while having 10% portfolio A2, potentially due to their lower performance during 2018-2022 (not reported in the paper). Coca-Cola, Pfizer, and Merck & Company contributed 0% in the A2 portfolio now to be weighted up at 4%, 7%, and 10% respectively, indicating better performance in the latter period.

Noticeably, portfolio B adds another energy company, whose energy sector sums up to 12% compared to only 10% in portfolio A2. Energy companies tend to have more volatility, depending on their level of return compensation. Generally, both two portfolios did not contain much weight from energy companies, potentially due to their low performance during the periods.

COMPANY	WEIGHT	
	PORTFOLIO A2	PORTFOLIO B
UNITEDHEALTH GROUP	10%	10%
INTERNATIONAL BUS.MCHS.	0%	0%
JOHNSON & JOHNSON	10%	0%
WALMART	10%	10%
JP MORGAN CHASE & CO.	0%	0%
P&G	0%	10%
VISA	10%	0%
ELI LILLY	10%	10%
HOME DEPOT	10%	0%
MASTERCARD	9.52%	10%
BANK OF AMERICA	8.44%	0%
COCA COLA	0%	4.63%
PFIZER	0%	7.47%
MERCK & COMPANY	2.04%	10%
MORGAN STANLEY	0%	0%
THERMO FISHER SCIENTIFIC	10%	10%
CITIGROUP	0%	5.59%

NEXTERA ENERGY	10%	10%
EXXON MOBIL	0%	2.32%
CHEVRON	0%	0%
Total	100%	100%

Figure 9. Portfolio Weights of A2 and B.

Conclusion & Recommendation

Overall, our portfolio constructed in 2017 performs quite well in terms of risk-adjusted returns both during the 2 periods, 2012-2017 & 2018-2022 against its relative benchmark. The portfolio of the 2017 period saw a higher annual return than the S&P500, reflected by a significant t-value. This also is supported by having a higher Sharpe ratio than its relative benchmark.

In the later period, the portfolio outperformance was measured by having a significant and positive Alpha, while 90% of its variance is explained by the market risk. Our portfolio also moves quite close to the movement of the benchmark, reflected by a significant Beta value of 0.82. This indicates the ability to generate more excess return while having a relatively lower risk than the market.

As a risk-taking investor, one is encouraged to invest in portfolio B during 2018-2022 instead of portfolio A2 since it yields a much better-adjusted return for a similar level of risk, approximately more than 2.5% annual return. Portfolio B gives more weight to various sectors, such as the Beverage and Retail sectors, which makes it more diversified. An additional energy company is also added to the portfolio. However, the analysis is somewhat limited as it did not take into consideration the benchmark's total return index (TRI). Having analyzed the TRI might have given the S&P 500 a higher score in terms of average return, as it accounts for both capital gains and cash distributions.

Reference

The Dangers of Over-Diversifying Your Portfolio. (2022, September 30). Investopedia.

<https://www.investopedia.com/investing/dangers-over-diversifying-your-portfolio/>

How is the Banking Sector for Value Investing? (2021, December 4). Investopedia.

<https://www.investopedia.com/ask/answers/041715/banking-sector-good-choice-value-investing.asp>

Investing In The Healthcare Sector. (2021, December 24). Investopedia.

<https://www.investopedia.com/articles/stocks/08/investing-in-healthcare.asp>

Market Capitalization: How Is It Calculated and What Does It Tell Investors? (2022, August

11). Investopedia. <https://www.investopedia.com/terms/m/marketcapitalization.asp>

Morningstar, Inc. (2022, December 8). *5 Strategies for Portfolio Withdrawals in 2022.*

<https://www.morningstar.com/>